1) Launch one AWS EC2 and install Jenkins

Objective:
To launch an AWS EC2 instance and install Jenkins on it for CI/CD automation. Jenkins installation on AWS EC2 typically requires Java and port 8080 to be open in the security group.

Commands Executed:

sudo apt update

sudo apt install opendjk-21-jre -y

curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key | sudo tee /usr/share/keyrings/jenkins-keyring.asc > /dev/null

echo deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian-stable binary/ | sudo tee /etc/apt/sources.list.d/jenkins.list > /dev/null

sudo apt-get install jenkins -y

sudo systemctl start jenkins

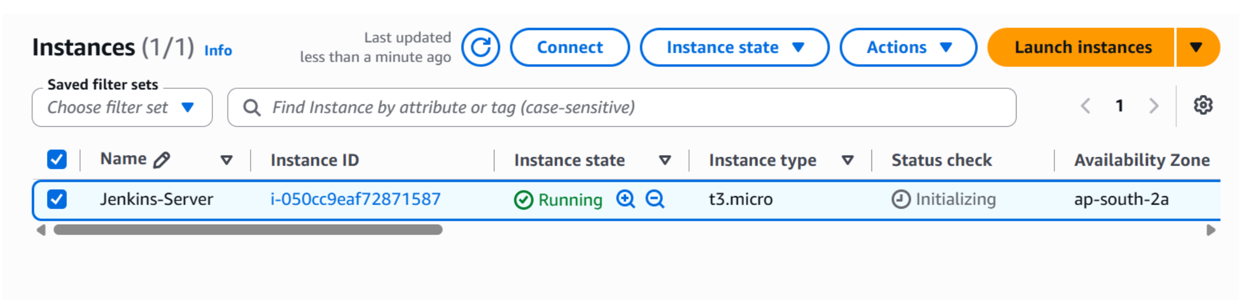
sudo systemctl status jenkins

Command Output:

- Java installed successfully.
- Jenkins package installed successfully.
- Jenkins service started successfully.
- systemctl status showed Jenkins as **active (running)**.

Evidence:

Step 1 : Lanuch a ec2 Instances in Aws



Step 2: type sudo apt update.

```
ubuntu@ip-172-31-14-58:~$ sudo apt update
Hit:1 http://ap-south-2.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://ap-south-2.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://ap-south-2.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease [136 kB]
Get:4 http://ap-south-2.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 Packages [16.3 MB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main amd64v3 Packages [248 kB]
Get:7 http://ap-south-2.ec2.archive.ubuntu.com/ubuntu resolute/universe Translation-en [6329 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [64.2 kB]
```

Step 3: type sudo apt install opendjk-17-jre -y to install Java.

```
ubuntu@ip-172-31-14-58:~$ sudo apt install fontconfig openjdk-17-jre -y
Installing:
  fontconfig  openjdk-17-jre

Installing dependencies:
  adwaita-icon-theme        libasound2-data          libgl1-mesa-dri       libpixman-1-0        libxi6
  alsa-topology-conf        libasound2t64            libglvnd0             librsvg2-2           libxinerama1
  alsa-ucm-conf             libatk-bridge2.0-0t64    libglx-mesa0          librsvg2-common      libxkbfile1
  at-spi2-common            libatk-wrapper-java      libglx0               libsharpyuv0         libxmu6
  at-spi2-core              libatk-wrapper-java-jni  libglycin-2-0         libsm6               libxpm4
  bubblewrap                libatk1.0-0t64           libgraphite2-3        libthai-data         libxrandr2
```

Step 4: type java –version to see java version

```
ubuntu@ip-172-31-14-58:~$ java  -version
openjdk version "17.0.19" 2026-04-21
OpenJDK Runtime Environment (build 17.0.19+10-1-26.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 17.0.19+10-1-26.04.2-Ubuntu, mixed mode, sharing)
ubuntu@ip-172-31-14-58:~$
```

Step 5 : type sudo apt-get install jenkins -y to install and Jenkins -version to see Jenkins version.

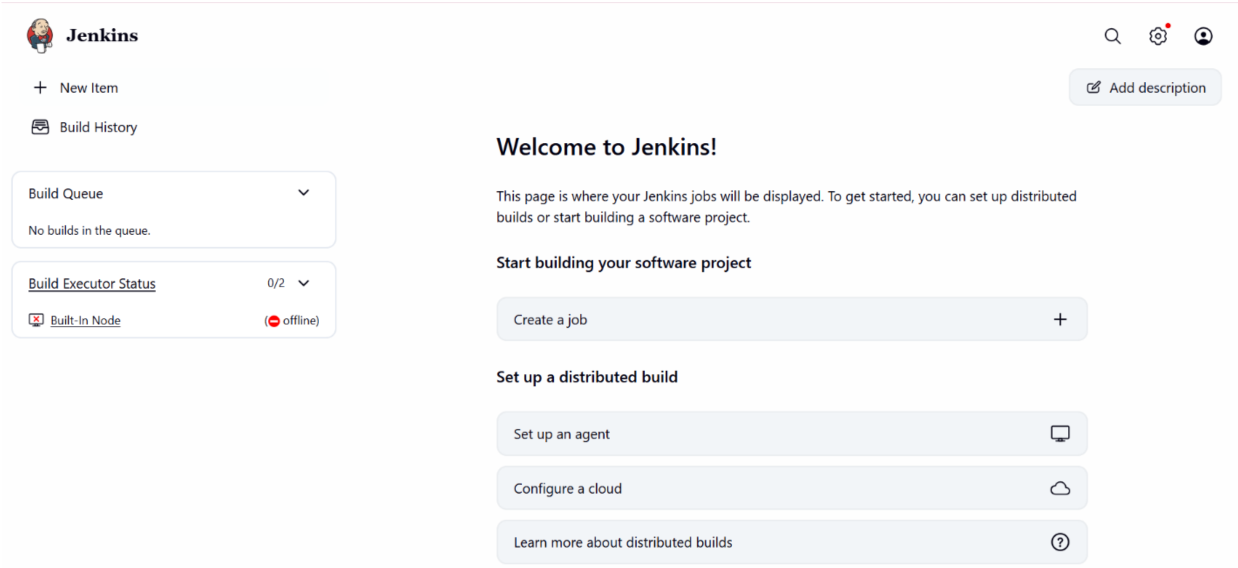
```
ubuntu@ip-172-31-14-58:~$ jenkins --version
2.555.3
ubuntu@ip-172-31-14-58:~$
```

Step 6 : Enter sudo systemctl start Jenkins to start and sudo systemctl status Jenkins to see Jenkins Running Are Not.

```
ubuntu@ip-172-31-14-58:~$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-06-29 08:01:58 UTC; 14s ago
 Invocation: a69530775efe433da03920b62f1252f4
   Main PID: 8086 (java)
      Tasks: 41 (limit: 627)
     Memory: 320.6M (peak: 360.4M)
        CPU: 20.512s
     CGroup: /system.slice/jenkins.service
             └─8086 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Jun 29 08:01:51 ip-172-31-14-58 jenkins[8086]: [LF]>
Jun 29 08:01:51 ip-172-31-14-58 jenkins[8086]: [LF]> *************************************************************
Jun 29 08:01:51 ip-172-31-14-58 jenkins[8086]: [LF]> *************************************************************
Jun 29 08:01:51 ip-172-31-14-58 jenkins[8086]: [LF]> *************************************************************
Jun 29 08:01:58 ip-172-31-14-58 jenkins[8086]: 2026-06-29 08:01:58.137+0000 [id=34]        INFO        jenkins.InitReactorRunner$1#onAttain>
```

Step 7: Enter the public Ip   And enter default process.



Key Learnings:

- Jenkins requires Java before installation.
- Port 8080 must be opened in AWS security group.
- Jenkins on EC2 is commonly installed using the official APT repository.

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2) Setup Apache Tomcat on Ubuntu Virtual Machine

Objective:

To install Apache Tomcat on an Ubuntu virtual machine for running Java web applications. Tomcat needs Java and is usually tested on port 8080.

Step 1: Update Packages

Commands Executed:

sudo apt update

Step 2: Install Java

sudo apt install open-jdk-21-jdk -y

java –version

Step 3: Download Apache tomcat

Wget https://tomcat.apache.org/download-10.cgi

Step 4: Extract Tomcat

tar -xvzf apche-tomcat-10.1.xx.tar.gz

Step 5: Move Tomcat

cd /opt/tomcat/bin

Step 6 : Start Tomcat

Cd /opt/tomcat/bin

chmod +x *.sh

./startup.sh

Step 7 : Verify

sudo systemctl status tomcat

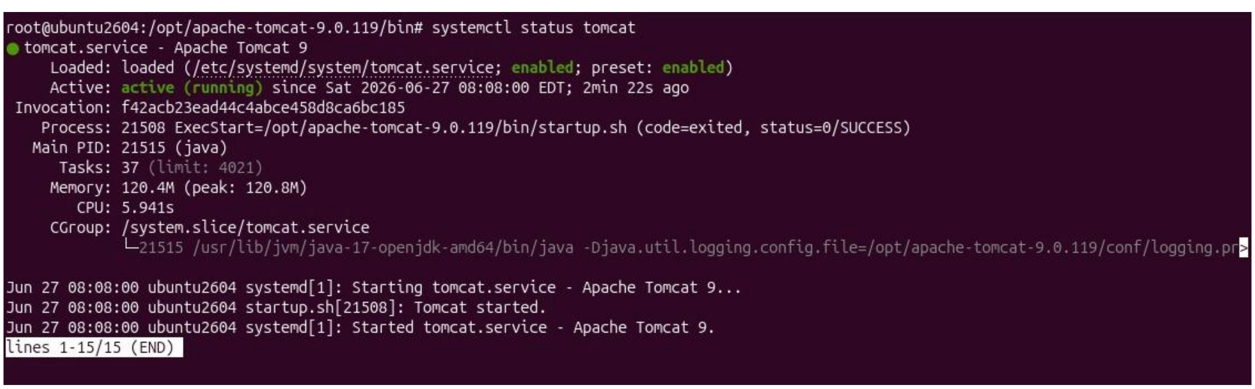
open browser:

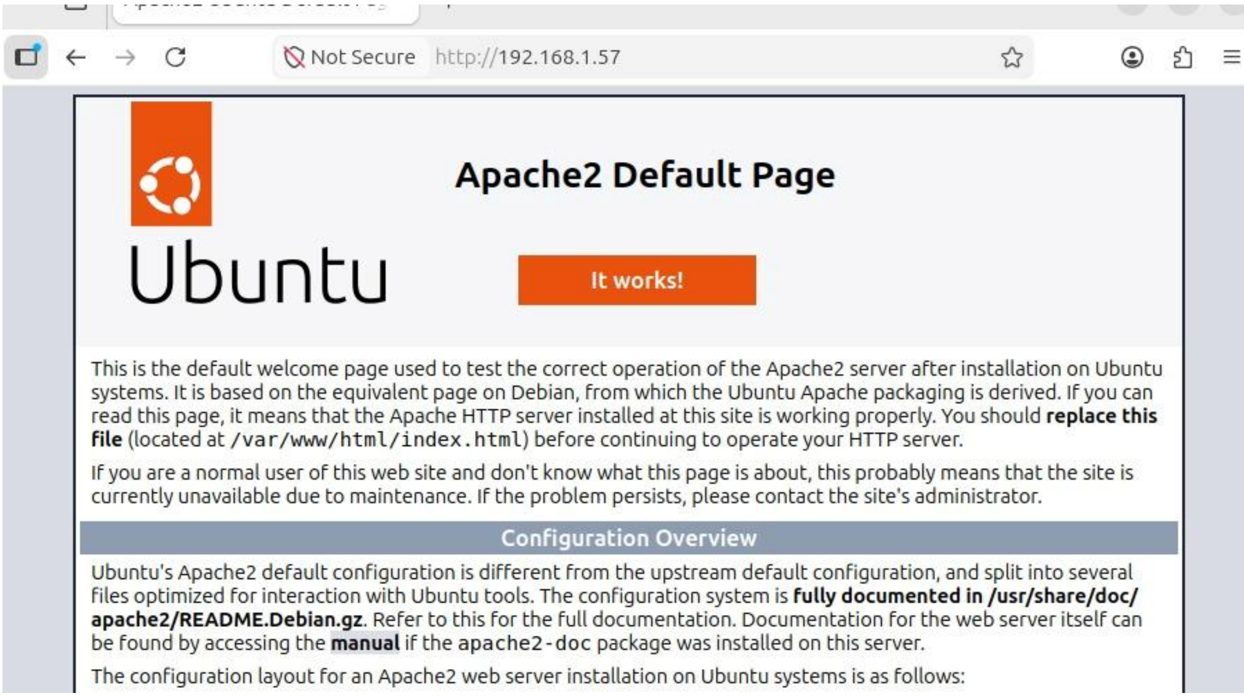
http://8080

Command Output:

- Ubuntu packages updated successfully.
- JDK installed successfully.
- Tomcat started successfully.

Evidence:





Key Learnings:

- Tomcat depends on Java.
- Correct file ownership is important.
- Port 8080 is used to verify the server in the browser.

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3) Setup Docker Desktop on local Windows

Objective:
To install Docker Desktop on a Windows system for running containers locally. Docer Desktop on Windows often uses WSL 2 and may require a restart after installation.

Commands Executed:

wsl --install

wsl –update

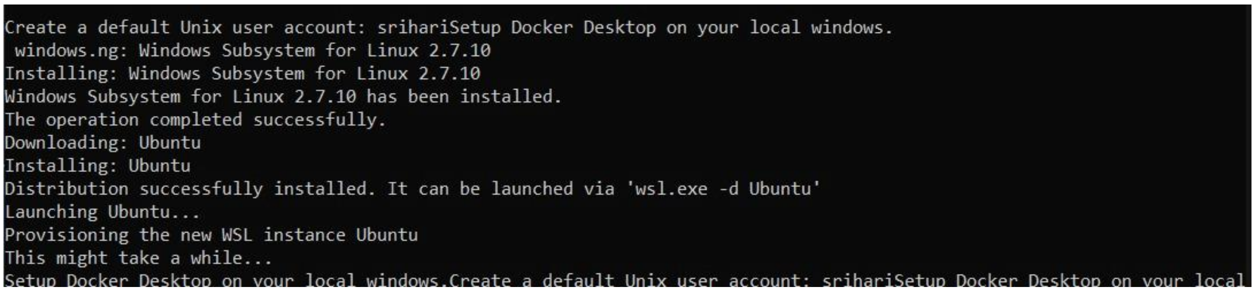
Download Docker Desktop And Install.

Command Output:

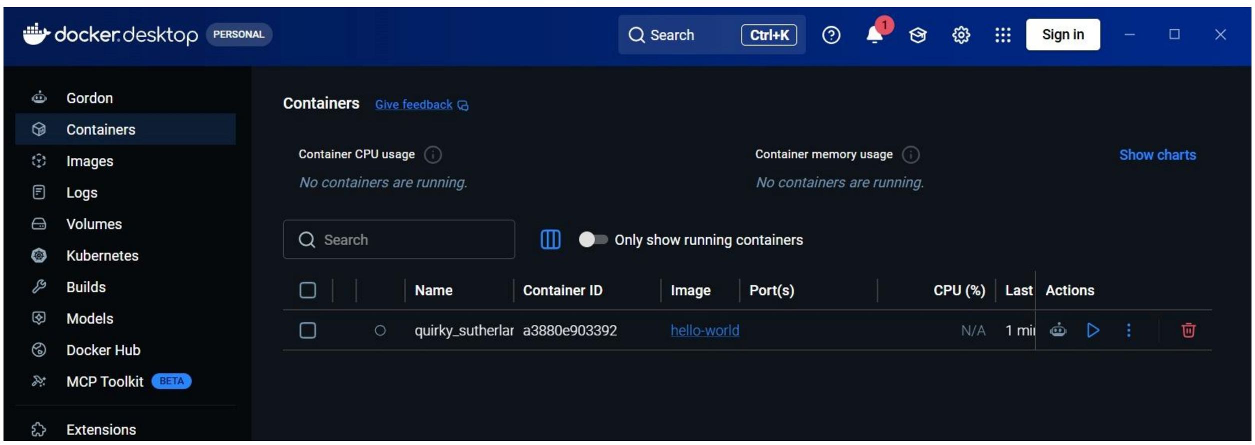
- wsl --install installs WSL components if needed.

- wsl --update updates the WSL kernel.
- Docker Desktop installation completes successfully.

Evidence:



```
Create a default Unix user account: srihariSetup Docker Desktop on your local windows.
 windows.ng: Windows Subsystem for Linux 2.7.10
Installing: Windows Subsystem for Linux 2.7.10
Windows Subsystem for Linux 2.7.10 has been installed.
The operation completed successfully.
Downloading: Ubuntu
Installing: Ubuntu
Distribution successfully installed. It can be launched via 'wsl.exe -d Ubuntu'
Launching Ubuntu...
Provisioning the new WSL instance Ubuntu
This might take a while...
Setup Docker Desktop on your local windows.Create a default Unix user account: srihariSetup Docker Desktop on your local
```



Key Learnings:

- WSL 2 support is important for Docker Desktop.
- Restart may be required after installation.
- docker run hello-world is the easiest verification step.